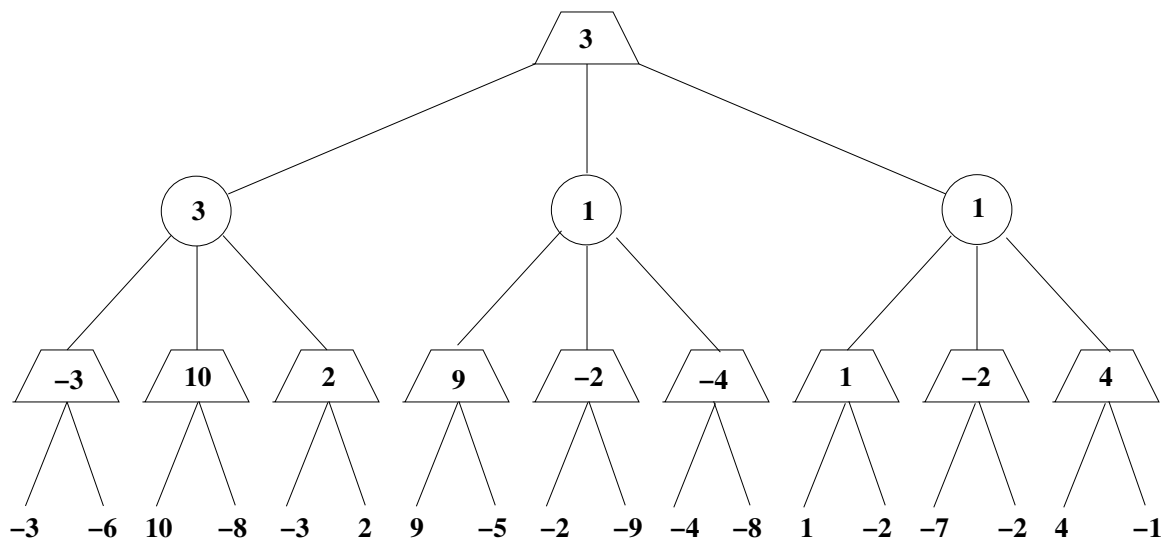


1 Expectimax

Consider the zero-sum expectimax game tree below. Circles represent chance nodes. Trapezoids that point up represent choices for the player seeking to maximize. Outcome values for the maximizing player are listed for each leaf node.

1. First, assume that each chance node chooses uniformly between available moves. Assuming optimal play, carry out the expectimax search algorithm and write the value of each node inside the corresponding trapezoid. What is the expected value of this game assuming optimal play? What is the optimal move to make at each node? Fill in the node values using a drawing program.



2 Probability

Consider the following probability tables for variables A, B, C .

A	$P(A)$	B	A	$P(B A)$	C	A	$P(C A)$
+a	0.3	+b	+a	0.7	+c	+a	0.2
-a	0.7	+b	-a	0.6	+c	-a	0.9

1. Derive $P(A, B)$ by filling in the entries below. What formula are you using?

The product formula $P(A, B) = P(B|A)P(A)$ is used below

A	B	$P(A, B)$
+a	+b	$0.3 \cdot 0.7 = 0.21$
+a	-b	$0.3 \cdot 0.3 = 0.09$
-a	+b	$0.7 \cdot 0.6 = 0.42$
-a	-b	$0.7 \cdot 0.4 = 0.28$

2. Derive $P(A|C)$ by filling in the entries below. What formula are you using?

Bayes' rule is used to change the causality: $P(A|C) = P(C|A)P(A)/P(C)$. First, find $P(A, C) = P(C|A)P(A)$ via the product rule.

A	C	$P(A, C)$
+a	+c	$0.3 \cdot 0.2 = 0.06$
+a	-c	$0.3 \cdot 0.8 = 0.24$
-a	+c	$0.7 \cdot 0.9 = 0.63$
-a	-c	$0.7 \cdot 0.1 = 0.07$

Next marginalize to find $P(C)$.

C	$P(C)$
+c	$0.06 + 0.63 = 0.69$
-c	$0.24 + 0.07 = 0.31$

Then apply Bayes' rule. The result is two different conditional probability tables.

A	C	$P(A C)$
+a	+c	$0.2 \cdot 0.3 / 0.69 = 0.087$
-a	+c	$0.9 \cdot 0.7 / 0.69 = 0.913$
+a	-c	$0.8 \cdot 0.3 / 0.31 = 0.774$
-a	-c	$0.1 \cdot 0.7 / 0.31 = 0.226$

3. Suppose B and C are conditionally independent given A . Derive the full joint distribution $P(A, B, C)$ by filling in the entries below. What is the formula you would use?

From the chain rule, $P(A, B, C) = P(A)P(B|A)P(C|A, B) = P(A)P(B|A)P(C|A)$.

A	B	C	$P(A, B, C)$
+a	+b	+c	$0.3*0.7*0.2=0.042$
+a	+b	-c	$0.3*0.7*0.8=0.168$
+a	-b	+c	$0.3*0.3*0.2=0.018$
+a	-b	-c	$0.3*0.3*0.8=0.072$
-a	+b	+c	$0.7*0.6*0.9=0.378$
-a	+b	-c	$0.7*0.6*0.1=0.042$
-a	-b	+c	$0.7*0.4*0.9=0.252$
-a	-b	-c	$0.7*0.4*0.1=0.028$

4. Use the full joint distribution you computed in problem 3 to derive the conditional distribution $P(A|+b, -c)$. What formula would you use? Then fill in the table below.

Use the product rule and the normalization trick: $P(A|+b, -c) = \alpha P(A, +b, -c)$. First, from the joint distribution,

A	B	C	$P(A, +b, -c)$
+a	+b	-c	0.168
-a	+b	-c	0.042

Then $\alpha = 1/(0.168 + 0.042) = 4.76$. Substituting,

A	B	C	$P(A +b, -c)$
+a	+b	-c	$4.76*0.168=0.800$
-a	+b	-c	$4.76*0.042=0.200$